Annapoorna S

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Summary

Systems Software Engineer with 3+ years at Hewlett Packard Enterprise building production-grade backend services and distributed automation for the world's largest HPC and AI clusters. Top 3 open-source contributor to HPE's flagship platform (1,600+ commits). Combines Python, Linux, Docker, Kubernetes, CI/CD, and Kafka with applied AI skills in LLM integration, RAG, and reinforcement learning.

Technical Skills

Languages: Python (advanced), C/C++, Java, SQL, Bash/Shell, TypeScript
AI/ML & LLM: LLM Integration (Llama 3.3, Groq), RAG Pipelines, Reinforcement Learning (PPO, CMDP), Scikit-learn, XGBoost, Pandas, NumPy, Feature Engineering, Gymnasium
Backend & Data: FastAPI, REST APIs, AsyncIO, Microservices, Distributed Systems, Kafka, PostgreSQL, MySQL
DevOps & Monitoring: Docker, Kubernetes, Podman, Jenkins, GitHub Actions, RPM Packaging, Artifactory, Git, Grafana, Prometheus
Practices: System Design, TDD (85%+ coverage), Code Reviews, Open Source, Agile/Scrum, Technical Documentation

Experience

Systems Software Engineer 2 – HPC & AI Feb 2025 – Present
Hewlett Packard Enterprise (HPE) *Bengaluru, India*

- Built a Python/FastAPI/AsyncIO **inventory service** and extended node collection with **Aruba management-switch collectors and parsers**; unified mixed-target CLI/API selection, collection and query services, and target-based results with corresponding tests
- Owned **NGSM** CMDB-driven add/delete/modify inventory workflows with asynchronous posting; improved node collection using **ClusterShell** batching, availability checks, and a PDSH fallback
- Evolved **HTT** diagnostics with reusable YAML-configured plugins, structured JSON output, and error-specific repair guidance; strengthened **SAT** automation through firmware-snapshot cleanup fixes and dependency vulnerability remediation

Systems Software Engineer – HPC & AI Sep 2023 – Feb 2025
Hewlett Packard Enterprise (HPE) *Bengaluru, India*

- Own end-to-end design and delivery of **System Admin Toolkit (SAT)**, a production CLI platform for boot/shutdown, firmware management, and configuration sync across 10,000+ node exascale clusters; top 3 open-source contributor with 1,600+ commits
- Designed and maintained **CI/CD pipelines** (Jenkins, GitHub Actions) across 6+ repositories; delivered 4 enterprise releases (HPE CSM v1.6–v1.7) with 85%+ test coverage through TDD
- Built **Hardware Triage Tool (HTT)**, an automated diagnostic engine with YAML-based workflows and Redfish API integration; covers 6 failure scenarios across 11 hardware platforms

R&D Software Engineer Intern Feb 2023 – Aug 2023
Hewlett Packard Enterprise (HPE) *Bengaluru, India*

- Built REST API clients, CI/CD pipelines (Jenkins, Git), and automation scripts; ramped up to independent feature delivery within 3 months

Projects

ThermoRL: RL for Thermal-Aware Workload Scheduling | *Python, PPO, Gymnasium* Research Prototype @ HPE

- Designed reinforcement learning scheduler co-optimizing compute placement across heterogeneous cooling zones (air, D2C, immersion) using Lagrangian-relaxed PPO with constrained MDP formulation; introduced Thermal Headroom Index (THI) as a unified zone-agnostic safety metric
- Achieved **73% reduction in thermal violations, 18% cooling energy savings, 12% carbon reduction** vs carbon-aware-only baselines in 50 MW AI campus digital-twin simulation

SAP O2C Graph Explorer | *FastAPI, React/TypeScript, PostgreSQL, LLM (Llama 3.3), RAG, Cytoscape.js* GitHub

- Full-stack AI application ingesting 19 SAP entity types and modeling Order-to-Cash flows as an interactive graph; LLM-powered chat interface with RAG architecture translates natural-language queries into live SQL execution
- Engineered LLM guardrails to constrain responses to dataset-relevant queries; deployed on Vercel + Render + Neon (PostgreSQL) with Docker containerization

Education

University of Texas at Austin 2025 – 2026
Post Graduate Program in Data Science & AI *GPA: 4.26 / 4.44*

JSS Science and Technology University (SJCE) 2019 – 2023
Bachelor of Engineering in Computer Science *CGPA: 9.25 / 10*

Achievements & Leadership

- Top-3 finisher in HPE internal Secured Code Tournament (twice); Product Owner for Inventory Service and SAT across 4 production releases; mentor at Nisvartha Foundation